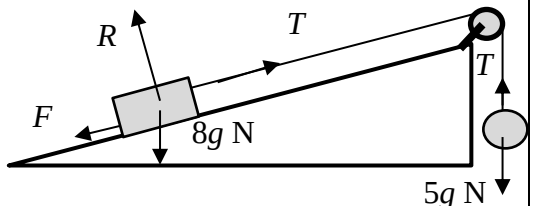
Question			Answer	Marks	AO	Guidance
13	(a)			B1 B1 B1 B1	1.1b 1.1b 1.1b 1.1b	(Each force must have correct direction) Matching tensions on the sphere and the block (may also include tensions on the pulley) Normal reaction. Allow without label or $8g\cos 15°$ used Friction in the correct direction and labelled This could be $0.3 \times 8g\cos 15°$ oe Both weights correct. Do not award if weight and its components shown together (unless dotted or similar) or any other extra forces
				[4]		
13	(b)		Sphere (downwards positive) $5g - T = 5a$	 M1 A1	 1.1a 1.1b	 Newton's second law allow sign errors All correct – accept $49 - T = 5a$ Any form
				[2]		
13	(c)		$R = 8g\cos 15°$ $F = 0.3R = 2.4g\cos 15°$ Newton's second law for the block $T - 8g\sin 15° - F = 8a$ Add equations of motion $5g - 8g\sin 15° - 2.4g\cos 15° = 13a$ $a = 0.461$	B1 M1 M1 A1 M1 A1	3.1b 3.4 1.1a 1.1b 1.1a 1.1b	 FT their R Allow one missing or incorrect force, must be dimensionally correct. FT their F All correct FT their F Attempt to eliminate T This equation with one missing or incorrect force implies the previous M1A0M1
				[6]		